Pengyuan Li, Ph.D.

Research Staff Member, MIT-IBM Watson AI Lab
Adjunct Faculty, Data Science Institute, University of Delaware

(302)-501-0291
pengyuan@ibm.com
pengyuanli

Professional Summary

Staff-level AI researcher and technical leader specializing in multimodal foundation models, document intelligence, and large-scale AI data systems. Lead developer of IBM Granite Vision Models, with end-to-end ownership across data acquisition, synthetic data generation, model training, evaluation, release, and enterprise-facing applications. Proven record of leading cross-functional research and engineering efforts, releasing impactful open-source models and datasets, mentoring researchers and interns, and translating frontier AI research into practical enterprise AI solutions.

Selected Highlights

Model Impact	Lead developer of **IBM Granite Vision Models**; 1M+ Hugging Face downloads, 1M+ Ollama uses, and 1M+ uses on IBM watsonx.ai platform.
Data Leadership	Led acquisition and processing of **10+ PB** of text, code, and multimodal data for IBM Granite language, code, and vision model development.
Benchmark Leadership	Led multimodal dataset and benchmark efforts for document intelligence, including **ChartNet** *(30k+ HF downloads)*, **ChartNet-Bench**, and **OmniDocBench-Pro**.
Technical Leadership	Led cross-functional collaborations across 3 internal teams, 20+ contributors, legal stakeholders, and external partners for multimodal data generation, filtering, annotation, and foundation model development.
Research Leadership	Organizer of **DataMFM Workshop** at CVPR 2026 and **AI and Biodata Resources Workshop** at Biocuration 2025
External Visibility	Invited speaker for *Granite Vision Models* at **OpenCV Live Talk**, reaching 24k+ YouTube views.

Awards & Honors

2022–2025	IBM Corporate Accomplishment Awards: A-Level, O-Level, and Special Accomplishment.
2026	Best Paper Award Nominee, ESWC 2026.
2021	Frank A. Pehrson Graduate Student Award, University of Delaware.
2020	Distinguished Graduate Student Award and Dissertation Fellowship.

Professional Service

Workshop Leadership	DataMFM Workshop and Challenge, CVPR 2026; AI and Biodata Resources Workshop, Biocuration 2025.
Area Chair	ACL 2026; EMNLP 2025–2026; BIBM 2020–2024.
Reviewer	ICML, NeurIPS, ACL, SIGKDD, SIGIR, WWW, ISMB, Bioinformatics, BIBM, etc.

Industry Experience

2021–Present **Research Staff Member**, *IBM Research*

- **Granite Vision Models:** Led development across data collection, filtering, synthetic data generation, model training, evaluation, release, and enterprise-facing document intelligence applications.

- **Document Intelligence Applications:** Helped develop the Granite Vision + Docling document intelligence demo showcased at IBM Think 2026, supporting use cases such as chart extraction, table understanding, and structured document analysis.
- **Synthetic Data Generation:** Led collaboration across 3 internal teams, 20+ contributors, and an external partner to create synthetic datasets for chart understanding, visual question answering, and data rewriting, improving Granite Vision performance.
- **Benchmark Creation:** Led benchmark design and dataset creation for chart and document intelligence, including ChartNet-Bench and OmniDocBench-Pro; coordinated with external partners such as Abaka AI on annotation, quality control, and benchmark construction.
- **Safe Multimodal Data Pipelines:** Oversaw preprocessing pipelines for multimodal data, including NSFW/CSAM detection and filtering; partnered with legal and cross-functional teams to support compliant VLM training. **IBM A-Level Accomplishment 2024**.
- **Foundation Model Data:** Led large-scale data acquisition, filtering, and processing for IBM Granite language, code, and multimodal model development. **IBM Special Accomplishment 2023; O-Level Accomplishment 2024**.
- **Enterprise AI Systems:** Built an enterprise search and recommendation system to match client requirements with IBM solutions. **IBM A-Level Accomplishment 2022**.
- **Mentorship:** Mentored 10+ summer interns on ML, VLMs, LLMs, and scientific document understanding, leading to publications and patent filings.

Jun–Aug 2019 **Research Intern**, *IBM Research*
- Applied NLP and topic modeling to customer reviews for identifying customers' pain points

Academic Experience

2023–Present **Collaborator**, *Sternberg Lab, Caltech*
- **Image Manipulation Detection in Scientific Literature:** Partnered with the microPublication Journal to design and deploy an image manipulation detection system in scientific submissions
- **LLM-Assisted Gene Summarization:** Collaborated with the Alliance of Genome Resources to develop a LLM-based method that accelerate gene summarization process

2023–Present **Adjunct Faculty**, *University of Delaware, Data Science Institute*
- Designed and taught BINF601: Introduction to Data Sciences, including biomedical image analysis and data science fundamentals

2015–2021 **Research Assistant**, *University of Delaware*
- Conducted Ph.D. research on biomedical document understanding, scientific figure extraction, compound figure separation, biomedical image classification, and multimodal health data analysis.

Education

2015–2021 **Ph.D., Computer Science**, *University of Delaware*

2011–2014 **M.E., Computer Software and Theory**, *Harbin Engineering University*

2007–2011 **B.E., Computer Science and Technology**, *Zhengzhou University*

Selected Publications

[1] *ChartNet: A Million-Scale, High-Quality Multimodal Dataset for Robust Chart Understanding*. CVPR 2026.

[2] *Context-Aware Visual Multi-Turn Conversation Generation from Wikipedia and Wikidata*. ESWC 2026. **(Best Paper Award Nominee)**

[3] *Composition-Grounded Instruction Synthesis for Visual Reasoning*. ICLR 2026.

...Full list available upon request or via Google Scholar.